

Student Management System — Frontend Source Code (React + Vite)

Single-document export of the full frontend codebase for the AWS Cloud & DevOps internship project.

package.json

```
{  
  
  "name": "sms-frontend",  
  
  "version": "1.0.0",  
  
  "description": "Student Management System - Frontend (React + Vite)",  
  
  "type": "module",  
  
  "scripts": {  
  
    "dev": "vite",  
  
    "build": "vite build",  
  
    "preview": "vite preview --host --port 4173"  
  
  },  
  
  "author": "Anil",  
  
  "license": "MIT",  
  
  "dependencies": {  
  
    "axios": "^1.7.7",  
  
    "react": "^18.3.1",  
  
    "react-dom": "^18.3.1",
```

```
    "react-router-dom": "^6.26.2"

  },

  "devDependencies": {

    "@vitejs/plugin-react": "^4.3.1",

    "vite": "^5.4.6"

  }

}
```

vite.config.js

```
import { defineConfig } from 'vite';

import react from '@vitejs/plugin-react';

// https://vitejs.dev/config/

export default defineConfig({

  plugins: [react()],

  server: {

    host: true,

    port: 3000

  },

  preview: {

    host: true,

    port: 4173

  }

})
```

```
});
```

index.html

```
<!doctype html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Student Management System</title>

  </head>

  <body>

    <div id="root"></div>

    <script type="module" src="/src/main.jsx"></script>

  </body>

</html>
```

.env.example

```
# URL of the backend API (no trailing slash). In docker-compose this points
# to the backend service name; in production it points at the AWS ALB /
# API Gateway endpoint and is baked in at build time by Vite.

VITE_API_BASE_URL=http://localhost:5000
```

src/index.css

```
@import
url('https://fonts.googleapis.com/css2?family=Sora:wght@500;600;700&family=Inter:wght@400;
500;600&family=IBM+Plex+Mono:wght@500&display=swap');

:root {

  --color-bg: #F5F7FA;

  --color-surface: #FFFFFF;

  --color-navy: #1E3A5F;

  --color-navy-dark: #16293F;

  --color-amber: #E8A33D;

  --color-text: #1F2430;

  --color-text-muted: #6B7280;

  --color-border: #E4E8EE;

  --color-success: #2F9E64;

  --color-success-bg: #E6F6ED;

  --color-danger: #D64550;

  --color-danger-bg: #FCEAEC;

  --color-info: #2C7BE5;

  --color-info-bg: #E9F2FE;

  --font-display: 'Sora', sans-serif;

  --font-body: 'Inter', sans-serif;

  --font-mono: 'IBM Plex Mono', monospace;
```

```
--radius: 8px;

--shadow-sm: 0 1px 2px rgba(16, 24, 40, 0.06);

--shadow-md: 0 4px 12px rgba(16, 24, 40, 0.08);

}

* { box-sizing: border-box; }

html, body, #root {

  height: 100%;

  margin: 0;

}

body {

  font-family: var(--font-body);

  background: var(--color-bg);

  color: var(--color-text);

  -webkit-font-smoothing: antialiased;

}

h1, h2, h3, h4 {

  font-family: var(--font-display);

  font-weight: 600;

  margin: 0;

  letter-spacing: -0.01em;

}

a { color: inherit; }
```

```
button, input, select, textarea {  
  
    font-family: var(--font-body);  
  
    font-size: 0.9rem;  
  
}  
  
button:focus-visible,  
  
input:focus-visible,  
  
select:focus-visible,  
  
a:focus-visible {  
  
    outline: 2px solid var(--color-info);  
  
    outline-offset: 2px;  
  
}  
  
/* ----- Layout shell ----- */  
  
.app-shell {  
  
    display: flex;  
  
    min-height: 100vh;  
  
}  
  
.sidebar {  
  
    width: 240px;  
  
    flex-shrink: 0;  
  
    background: var(--color-navy-dark);  
  
    color: #E7ECF3;
```

```
display: flex;

flex-direction: column;

padding: 24px 0;
}

.sidebar-brand {

font-family: var(--font-display);

font-size: 1.05rem;

font-weight: 700;

padding: 0 24px 24px;

border-bottom: 1px solid rgba(255,255,255,0.08);

margin-bottom: 12px;

display: flex;

align-items: center;

gap: 8px;
}

.sidebar-brand .dot { color: var(--color-amber); }

.sidebar-nav { display: flex; flex-direction: column; gap: 2px; padding: 0 12px; }

.sidebar-link {

display: flex;

align-items: center;

gap: 10px;

padding: 10px 12px;
```

```
border-radius: var(--radius);

text-decoration: none;

color: #C4CEDB;

font-size: 0.9rem;

font-weight: 500;

transition: background 0.15s ease, color 0.15s ease;
}

.sidebar-link:hover { background: rgba(255,255,255,0.06); color: #fff; }

.sidebar-link.active { background: var(--color-navy); color: #fff; }

.sidebar-footer {

margin-top: auto;

padding: 16px 24px 0;

border-top: 1px solid rgba(255,255,255,0.08);

font-size: 0.8rem;

color: #93A2B5;

}

.main-area { flex: 1; display: flex; flex-direction: column; min-width: 0; }

.topbar {

height: 64px;

background: var(--color-surface);

border-bottom: 1px solid var(--color-border);
```



```
display: flex;

align-items: center;

justify-content: space-between;

padding: 0 28px;
}

.topbar h1 { font-size: 1.15rem; }

.user-chip {

display: flex;

align-items: center;

gap: 10px;

font-size: 0.85rem;
}

.user-avatar {

width: 32px; height: 32px;

border-radius: 50%;

background: var(--color-navy);

color: #fff;

display: flex; align-items: center; justify-content: center;

font-family: var(--font-display);

font-weight: 600;

font-size: 0.8rem;
}
```

```
.logout-btn {  
  
  border: 1px solid var(--color-border);  
  
  background: transparent;  
  
  padding: 6px 12px;  
  
  border-radius: var(--radius);  
  
  cursor: pointer;  
  
  color: var(--color-text-muted);  
  
}  
  
.logout-btn:hover { border-color: var(--color-danger); color: var(--color-danger); }  
  
.page-content { padding: 28px; flex: 1; }  
  
/* ----- Cards & tables ----- */  
  
.card {  
  
  background: var(--color-surface);  
  
  border: 1px solid var(--color-border);  
  
  border-radius: 12px;  
  
  box-shadow: var(--shadow-sm);  
  
}  
  
.card-header {  
  
  display: flex;  
  
  align-items: center;  
  
  justify-content: space-between;
```

```
padding: 18px 20px;

border-bottom: 1px solid var(--color-border);
}

.stat-grid {

display: grid;

grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));

gap: 16px;

margin-bottom: 24px;
}

.stat-card {

background: var(--color-surface);

border: 1px solid var(--color-border);

border-radius: 12px;

padding: 18px 20px;

box-shadow: var(--shadow-sm);
}

.stat-card .stat-label {

font-size: 0.78rem;

color: var(--color-text-muted);

text-transform: uppercase;

letter-spacing: 0.04em;

font-weight: 600;
```

```
}
```

```
.stat-card .stat-value {
```

```
  font-family: var(--font-display);
```

```
  font-size: 1.9rem;
```

```
  margin-top: 6px;
```

```
  color: var(--color-navy);
```

```
}
```

```
table { width: 100%; border-collapse: collapse; }
```

```
thead th {
```

```
  text-align: left;
```

```
  font-size: 0.75rem;
```

```
  text-transform: uppercase;
```

```
  letter-spacing: 0.04em;
```

```
  color: var(--color-text-muted);
```

```
  padding: 12px 20px;
```

```
  border-bottom: 1px solid var(--color-border);
```

```
  background: #FAFBFC;
```

```
}
```

```
tbody td {
```

```
  padding: 14px 20px;
```

```
  border-bottom: 1px solid var(--color-border);
```

```
font-size: 0.9rem;

}

tbody tr:last-child td { border-bottom: none; }

tbody tr:hover { background: #FAFBFD; }

.roll-badge {

font-family: var(--font-mono);

font-size: 0.78rem;

background: #EEF2F7;

padding: 3px 8px;

border-radius: 6px;

color: var(--color-navy);

}

.status-pill {

display: inline-flex;

align-items: center;

gap: 5px;

font-size: 0.75rem;

font-weight: 600;

padding: 3px 10px;

border-radius: 999px;

}

.status-pill.active { background: var(--color-success-bg); color: var(--color-success); }
```

```
.status-pill.inactive { background: var(--color-danger-bg); color: var(--color-danger); }

.status-pill.graduated { background: var(--color-info-bg); color: var(--color-info); }

.row-actions { display: flex; gap: 8px; }

/* ----- Buttons ----- */

.btn {

  display: inline-flex;

  align-items: center;

  gap: 6px;

  padding: 8px 16px;

  border-radius: var(--radius);

  border: 1px solid transparent;

  font-weight: 600;

  font-size: 0.85rem;

  cursor: pointer;

  transition: all 0.15s ease;

}

.btn-primary { background: var(--color-navy); color: #fff; }

.btn-primary:hover { background: var(--color-navy-dark); }

.btn-accent { background: var(--color-amber); color: #3A2A0C; }

.btn-accent:hover { filter: brightness(0.95); }

.btn-ghost { background: transparent; border-color: var(--color-border); color: var(--color-text); }
```

```
.btn-ghost:hover { border-color: var(--color-navy); }

.btn-danger-ghost { background: transparent; border-color: var(--color-border); color:
var(--color-danger); }

.btn-danger-ghost:hover { background: var(--color-danger-bg); }

.btn-sm { padding: 5px 10px; font-size: 0.78rem; }

.btn-block { width: 100%; justify-content: center; }

/* ----- Forms ----- */

.form-group { margin-bottom: 16px; }

.form-group label {

    display: block;

    font-size: 0.82rem;

    font-weight: 600;

    margin-bottom: 6px;

    color: var(--color-text);

}

.form-control {

    width: 100%;

    padding: 9px 12px;

    border: 1px solid var(--color-border);

    border-radius: var(--radius);

    background: #fff;

    color: var(--color-text);
```

```
}
```

```
.form-control:focus { border-color: var(--color-navy); }
```

```
.form-row { display: grid; grid-template-columns: 1fr 1fr; gap: 16px; }
```

```
.search-input {
```

```
  border: 1px solid var(--color-border);
```

```
  border-radius: var(--radius);
```

```
  padding: 8px 12px;
```

```
  min-width: 260px;
```

```
  background: #fff;
```

```
}
```

```
.alert {
```

```
  padding: 10px 14px;
```

```
  border-radius: var(--radius);
```

```
  font-size: 0.85rem;
```

```
  margin-bottom: 16px;
```

```
}
```

```
.alert-danger { background: var(--color-danger-bg); color: var(--color-danger); }
```

```
.alert-success { background: var(--color-success-bg); color: var(--color-success); }
```

```
.empty-state {
```

```
  text-align: center;
```

```
  padding: 48px 20px;
```

```
  color: var(--color-text-muted);
```



```
}
```

```
/* ----- Auth pages ----- */
```

```
.auth-wrapper {
```

```
  min-height: 100vh;
```

```
  display: flex;
```

```
  align-items: center;
```

```
  justify-content: center;
```

```
  background: linear-gradient(135deg, var(--color-navy-dark) 0%, var(--color-navy) 55%,  
#2A5688 100%);
```

```
}
```

```
.auth-card {
```

```
  width: 380px;
```

```
  background: var(--color-surface);
```

```
  border-radius: 14px;
```

```
  padding: 36px 32px;
```

```
  box-shadow: var(--shadow-md);
```

```
}
```

```
.auth-brand {
```

```
  font-family: var(--font-display);
```

```
  font-size: 1.3rem;
```

```
  font-weight: 700;
```

```
  margin-bottom: 4px;
```

```
color: var(--color-navy);

}

.auth-sub { color: var(--color-text-muted); font-size: 0.85rem; margin-bottom: 24px; }

.auth-switch { text-align: center; margin-top: 16px; font-size: 0.85rem; color:
var(--color-text-muted); }

.auth-switch a { color: var(--color-navy); font-weight: 600; text-decoration: none; }

.modal-backdrop {

position: fixed; inset: 0;

background: rgba(16, 24, 40, 0.45);

display: flex; align-items: center; justify-content: center;

z-index: 50;

}

.modal-panel {

width: 480px;

max-width: 92vw;

max-height: 88vh;

overflow-y: auto;

background: #fff;

border-radius: 12px;

padding: 24px;

box-shadow: var(--shadow-md);

}
```

```
.modal-panel h2 { margin-bottom: 18px; font-size: 1.1rem; }
```

```
.pagination { display: flex; gap: 8px; align-items: center; justify-content: flex-end; padding: 16px 20px; }
```

src/main.jsx

```
import React from 'react';

import ReactDOM from 'react-dom/client';

import App from './App.jsx';

import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(

  <React.StrictMode>

    <App />

  </React.StrictMode>

);
```

src/App.jsx

```
import { BrowserRouter, Routes, Route } from 'react-router-dom';

import { AuthProvider } from './context/AuthContext';

import ProtectedRoute from './components/ProtectedRoute';

import Layout from './components/Layout';

import Login from './pages/Login';

import Register from './pages/Register';

import Dashboard from './pages/Dashboard';
```

```
import Students from './pages/Students';

import Courses from './pages/Courses';

import Enrollments from './pages/Enrollments';

export default function App() {

  return (

    <BrowserRouter>

      <AuthProvider>

        <Routes>

          <Route path="/login" element={<Login />} />

          <Route path="/register" element={<Register />} />

          <Route

            path="/"

            element={

              <ProtectedRoute>

                <Layout />

              </ProtectedRoute>

            }

          >

            <Route index element={<Dashboard />} />

            <Route path="students" element={<Students />} />

            <Route path="courses" element={<Courses />} />

          </Route>

        </Routes>

      </AuthProvider>

    </BrowserRouter>

  );
}
```

```

        <Route path="enrollments" element={<Enrollments />} />

    </Route>

</Routes>

</AuthProvider>

</BrowserRouter>

);

}

```

src/context/AuthContext.jsx

```

import { createContext, useContext, useState, useEffect } from 'react';

import { authService } from '../services/authService';

const AuthContext = createContext(null);

export function AuthProvider({ children }) {

    const [user, setUser] = useState(null);

    const [loading, setLoading] = useState(true);

    useEffect(() => {

        const storedUser = localStorage.getItem('sms_user');

        const token = localStorage.getItem('sms_token');

        if (storedUser && token) {

            setUser(JSON.parse(storedUser));

        }

        setLoading(false);
    });

```

```
}, []);
```

```
const login = async (email, password) => {
```

```
  const { data } = await authService.login(email, password);
```

```
  localStorage.setItem('sms_token', data.token);
```

```
  localStorage.setItem('sms_user', JSON.stringify(data.user));
```

```
  setUser(data.user);
```

```
  return data.user;
```

```
};
```

```
const register = async (payload) => {
```

```
  const { data } = await authService.register(payload);
```

```
  localStorage.setItem('sms_token', data.token);
```

```
  localStorage.setItem('sms_user', JSON.stringify(data.user));
```

```
  setUser(data.user);
```

```
  return data.user;
```

```
};
```

```
const logout = () => {
```

```
  localStorage.removeItem('sms_token');
```

```
  localStorage.removeItem('sms_user');
```

```
  setUser(null);
```

```
};
```

```
return (
```

```
  <AuthContext.Provider value={{ user, loading, login, register, logout }}>
```

```
    {children}

    </AuthContext.Provider>

  );
}

export function useAuth() {

  return useContext(AuthContext);

}
```

src/components/ProtectedRoute.jsx

```
import { Navigate } from 'react-router-dom';

import { useAuth } from '../context/AuthContext';

export default function ProtectedRoute({ children }) {

  const { user, loading } = useAuth();

  if (loading) return null;

  if (!user) return <Navigate to="/login" replace />;

  return children;

}
```

src/components/Layout.jsx

```
import { NavLink, Outlet, useNavigate } from 'react-router-dom';

import { useAuth } from '../context/AuthContext';

const NAV_ITEMS = [
```

```
{ to: '/', label: 'Dashboard', end: true },  
  
{ to: '/students', label: 'Students' },  
  
{ to: '/courses', label: 'Courses' },  
  
{ to: '/enrollments', label: 'Enrollments' }  
  
];
```

```
export default function Layout() {  
  
  const { user, logout } = useAuth();  
  
  const navigate = useNavigate();  
  
  const handleLogout = () => {  
  
    logout();  
  
    navigate('/login');  
  
  };  
  
  const initials = (user?.name || '?')  
  
    .split(' ')  
  
    .map((p) => p[0])  
  
    .join("")  
  
    .slice(0, 2)  
  
    .toUpperCase();  
  
  return (  
  
    <div className="app-shell">  
  
      <aside className="sidebar">  
  
        <div className="sidebar-brand">
```



```
<span>Campus</span><span className="dot">SMS</span>
```

```
</div>
```

```
<nav className="sidebar-nav">
```

```
{NAV_ITEMS.map((item) => (
```

```
<NavLink
```

```
key={item.to}
```

```
to={item.to}
```

```
end={item.end}
```

```
className={({ isActive }) => `sidebar-link${isActive ? ' active' : ''}`}
```

```
>
```

```
{item.label}
```

```
</NavLink>
```

```
)))
```

```
</nav>
```

```
<div className="sidebar-footer">
```

```
Student Management System
```

```
<br />
```

```
AWS Cloud & DevOps Internship
```

```
</div>
```

```
</aside>
```

```
<div className="main-area">
```

```
<header className="topbar">

  <h1>Student Management System</h1>

  <div className="user-chip">

    <div className="user-avatar">{initials}</div>

    <div>

      <div style={{ fontWeight: 600 }}>{user?.name}</div>

      <div style={{ color: 'var(--color-text-muted)', fontSize: '0.75rem', textTransform:
'capitalize' }}>

        {user?.role}

      </div>

    </div>

    <button className="logout-btn" onClick={handleLogout}>Log out</button>

  </div>

</header>

<main className="page-content">

  <Outlet />

</main>

</div>

</div>

);

}
```

src/components/ConfirmDialog.jsx

```
export default function ConfirmDialog({ title, message, onConfirm, onCancel }) {

  return (

    <div className="modal-backdrop" role="dialog" aria-modal="true">

      <div className="modal-panel" style={{ width: 360 }}>

        <h2>{title}</h2>

        <p style={{ color: 'var(--color-text-muted)', fontSize: '0.9rem', marginBottom: 20 }}>{message}</p>

        <div style={{ display: 'flex', gap: 10, justifyContent: 'flex-end' }}>

          <button className="btn btn-ghost" onClick={onCancel}>Cancel</button>

          <button className="btn btn-primary" style={{ background: 'var(--color-danger)' }}
onClick={onConfirm}>

            Delete

          </button>

        </div>

      </div>

    </div>

  );
}
```

src/components/StudentFormModal.jsx

```
import { useState } from 'react';

const EMPTY = {
```

```
rollNumber: "",
name: "",
email: "",
phone: "",
dateOfBirth: "",
address: "",
status: 'active',
courseId: ""
};

export default function StudentFormModal({ courses, initialData, onSave, onClose }) {

  const [form, setForm] = useState(initialData ? { ...EMPTY, ...initialData } : EMPTY);

  const [error, setError] = useState("");

  const [saving, setSaving] = useState(false);

  const handleChange = (e) => {

    const { name, value } = e.target;

    setForm((prev) => ({ ...prev, [name]: value }));

  };

  const handleSubmit = async (e) => {

    e.preventDefault();

    setError("");

    setSaving(true);
```

```

try {

  const payload = { ...form, courseId: form.courseId || null };

  await onSave(payload);

} catch (err) {

  setError(err.response?.data?.message || 'Failed to save student');

} finally {

  setSaving(false);

}

};

return (

  <div className="modal-backdrop" role="dialog" aria-modal="true">

    <div className="modal-panel">

      <h2>{initialData ? 'Edit Student' : 'Add Student'}</h2>

      {error && <div className="alert alert-danger">{error}</div>}

      <form onSubmit={handleSubmit}>

        <div className="form-row">

          <div className="form-group">

            <label htmlFor="rollNumber">Roll Number</label>

            <input

              id="rollNumber"

              name="rollNumber"

              className="form-control"

```

```
        value={form.rollNumber}

        onChange={handleChange}

        required

        disabled={!initialData}

    />

</div>

<div className="form-group">

    <label htmlFor="status">Status</label>

    <select id="status" name="status" className="form-control" value={form.status}
onChange={handleChange}>

        <option value="active">Active</option>

        <option value="inactive">Inactive</option>

        <option value="graduated">Graduated</option>

    </select>

</div>

</div>

<div className="form-group">

    <label htmlFor="name">Full Name</label>

    <input id="name" name="name" className="form-control" value={form.name}
onChange={handleChange} required />

</div>

<div className="form-group">
```

```
<label htmlFor="email">Email</label>
```

```
<input
```

```
  id="email"
```

```
  name="email"
```

```
  type="email"
```

```
  className="form-control"
```

```
  value={form.email}
```

```
  onChange={handleChange}
```

```
  required
```

```
</div>
```

```
<div className="form-row">
```

```
  <div className="form-group">
```

```
    <label htmlFor="phone">Phone</label>
```

```
    <input id="phone" name="phone" className="form-control" value={form.phone || ""}  
    onChange={handleChange} />
```

```
  </div>
```

```
  <div className="form-group">
```

```
    <label htmlFor="dateOfBirth">Date of Birth</label>
```

```
    <input
```

```
      id="dateOfBirth"
```

```
      name="dateOfBirth"
```

```
        type="date"

        className="form-control"

        value={form.dateOfBirth || ""}

        onChange={handleChange}

    />

</div>

</div>

<div className="form-group">

    <label htmlFor="address">Address</label>

    <input id="address" name="address" className="form-control" value={form.address || ""} onChange={handleChange} />

</div>

<div className="form-group">

    <label htmlFor="courseId">Course</label>

    <select id="courseId" name="courseId" className="form-control" value={form.courseId || ""} onChange={handleChange}>

        <option value="">Unassigned</option>

        {courses.map((c) => (

            <option key={c.id} value={c.id}>{c.code} — {c.name}</option>

        ))}

    </select>

</div>

<div style={{ display: 'flex', gap: 10, justifyContent: 'flex-end', marginTop: 8 }}>
```



```

        <button type="button" className="btn btn-ghost" onClick={onClose}>Cancel</button>

        <button type="submit" className="btn btn-primary" disabled={saving}>

            {saving ? 'Saving...' : 'Save Student'}

        </button>

    </div>

</form>

</div>

</div>

);
}

```

src/components/CourseFormModal.jsx

```

import { useState } from 'react';

const EMPTY = { code: "", name: "", durationMonths: 12, description: "" };

export default function CourseFormModal({ initialData, onSave, onClose }) {

    const [form, setForm] = useState(initialData ? { ...EMPTY, ...initialData } : EMPTY);

    const [error, setError] = useState("");

    const [saving, setSaving] = useState(false);

    const handleChange = (e) => {

        const { name, value } = e.target;

        setForm((prev) => ({ ...prev, [name]: value }));

    };

```

```

const handleSubmit = async (e) => {

  e.preventDefault();

  setError("");

  setSaving(true);

  try {

    await onSave({ ...form, durationMonths: Number(form.durationMonths) });

  } catch (err) {

    setError(err.response?.data?.message || 'Failed to save course');

  } finally {

    setSaving(false);

  }

};

return (

  <div className="modal-backdrop" role="dialog" aria-modal="true">

    <div className="modal-panel">

      <h2>{initialData ? 'Edit Course' : 'Add Course'}</h2>

      {error && <div className="alert alert-danger">{error}</div>}

      <form onSubmit={handleSubmit}>

        <div className="form-row">

          <div className="form-group">

            <label htmlFor="code">Course Code</label>

```

```
      <input id="code" name="code" className="form-control" value={form.code}
onChange={handleChange} required disabled={!initialData} />
```

```
    </div>
```

```
    <div className="form-group">
```

```
      <label htmlFor="durationMonths">Duration (months)</label>
```

```
      <input
```

```
        id="durationMonths"
```

```
        name="durationMonths"
```

```
        type="number"
```

```
        min="1"
```

```
        className="form-control"
```

```
        value={form.durationMonths}
```

```
        onChange={handleChange}
```

```
        required
```

```
      />
```

```
    </div>
```

```
</div>
```

```
<div className="form-group">
```

```
  <label htmlFor="name">Course Name</label>
```

```
  <input id="name" name="name" className="form-control" value={form.name}
onChange={handleChange} required />
```

```
</div>
```

```
<div className="form-group">
```

```
<label htmlFor="description">Description</label>

<textarea

  id="description"

  name="description"

  className="form-control"

  rows={3}

  value={form.description || ""}

  onChange={handleChange}

/>

</div>

<div style={{ display: 'flex', gap: 10, justifyContent: 'flex-end', marginTop: 8 }}>

  <button type="button" className="btn btn-ghost" onClick={onClose}>Cancel</button>

  <button type="submit" className="btn btn-primary" disabled={saving}>

    {saving ? 'Saving...' : 'Save Course'}

  </button>

</div>

</form>

</div>

</div>

);

}
```

src/components/EnrollmentFormModal.jsx

```
import { useState } from 'react';

export default function EnrollmentFormModal({ students, courses, onSave, onClose }) {

  const [form, setForm] = useState({ studentId: "", courseId: "", grade: "" });

  const [error, setError] = useState("");

  const [saving, setSaving] = useState(false);

  const handleChange = (e) => setForm({ ...form, [e.target.name]: e.target.value });

  const handleSubmit = async (e) => {

    e.preventDefault();

    setError("");

    setSaving(true);

    try {

      await onSave(form);

    } catch (err) {

      setError(err.response?.data?.message || 'Failed to create enrollment');

    } finally {

      setSaving(false);

    }

  };

  return (

    <div className="modal-backdrop" role="dialog" aria-modal="true">
```

```
<div className="modal-panel">

  <h2>New Enrollment</h2>

  {error && <div className="alert alert-danger">{error}</div>}

  <form onSubmit={handleSubmit}>

    <div className="form-group">

      <label htmlFor="studentId">Student</label>

      <select id="studentId" name="studentId" className="form-control"
value={form.studentId} onChange={handleChange} required>

        <option value="">Select a student</option>

        {students.map((s) => (

          <option key={s.id} value={s.id}>{s.rollNumber} — {s.name}</option>

        ))}

      </select>

    </div>

    <div className="form-group">

      <label htmlFor="courseId">Course</label>

      <select id="courseId" name="courseId" className="form-control" value={form.courseId}
onChange={handleChange} required>

        <option value="">Select a course</option>

        {courses.map((c) => (

          <option key={c.id} value={c.id}>{c.code} — {c.name}</option>

        ))}

      </select>

  </form>

</div>
```

```

    </div>

    <div className="form-group">

      <label htmlFor="grade">Grade (optional)</label>

      <input id="grade" name="grade" className="form-control" value={form.grade}
onChange={handleChange} />

    </div>

    <div style={{ display: 'flex', gap: 10, justifyContent: 'flex-end', marginTop: 8 }}>

      <button type="button" className="btn btn-ghost" onClick={onClose}>Cancel</button>

      <button type="submit" className="btn btn-primary" disabled={saving}>

        {saving ? 'Saving...' : 'Enroll Student'}

      </button>

    </div>

  </form>

</div>

</div>

);
}

```

src/services/api.js

```

import axios from 'axios';

// Base URL is injected at build/deploy time. In docker-compose and

// Kubernetes this is set via an environment variable so the same built

```

```
// bundle can point at different backend endpoints per environment.

const API_BASE_URL = import.meta.env.VITE_API_BASE_URL || 'http://localhost:5000';

const api = axios.create({

  baseURL: `${API_BASE_URL}/api`

});

api.interceptors.request.use((config) => {

  const token = localStorage.getItem('sms_token');

  if (token) {

    config.headers.Authorization = `Bearer ${token}`;

  }

  return config;

});

api.interceptors.response.use(

  (response) => response,

  (error) => {

    if (error.response && error.response.status === 401) {

      localStorage.removeItem('sms_token');

      localStorage.removeItem('sms_user');

      if (window.location.pathname !== '/login') {

        window.location.href = '/login';

      }

    }

  }

});
```



```
    return Promise.reject(error);

  }

);

export default api;
```

src/services/authService.js

```
import api from './api';

export const authService = {

  login: (email, password) => api.post('/auth/login', { email, password }),

  register: (payload) => api.post('/auth/register', payload),

  me: () => api.get('/auth/me')

};
```

src/services/studentService.js

```
import api from './api';

export const studentService = {

  list: (params) => api.get('/students', { params }),

  get: (id) => api.get(`/students/${id}`),

  create: (data) => api.post('/students', data),

  update: (id, data) => api.put(`/students/${id}`, data),

  remove: (id) => api.delete(`/students/${id}`)

};
```

src/services/courseService.js

```
import api from './api';

export const courseService = {

  list: () => api.get('/courses'),

  get: (id) => api.get(`/courses/${id}`),

  create: (data) => api.post('/courses', data),

  update: (id, data) => api.put(`/courses/${id}`, data),

  remove: (id) => api.delete(`/courses/${id}`)

};
```

src/services/enrollmentService.js

```
import api from './api';

export const enrollmentService = {

  list: () => api.get('/enrollments'),

  create: (data) => api.post('/enrollments', data),

  update: (id, data) => api.put(`/enrollments/${id}`, data),

  remove: (id) => api.delete(`/enrollments/${id}`)

};
```

src/pages/Login.jsx

```
import { useState } from 'react';

import { Link, useNavigate } from 'react-router-dom';
```

```
import { useAuth } from '../context/AuthContext';

export default function Login() {

  const [email, setEmail] = useState("");

  const [password, setPassword] = useState("");

  const [error, setError] = useState("");

  const [loading, setLoading] = useState(false);

  const { login } = useAuth();

  const navigate = useNavigate();

  const handleSubmit = async (e) => {

    e.preventDefault();

    setError("");

    setLoading(true);

    try {

      await login(email, password);

      navigate('/');

    } catch (err) {

      setError(err.response?.data?.message || 'Login failed. Check your credentials.');
```



```
    } finally {

      setLoading(false);

    }

  };

  return (
```

```
<div className="auth-wrapper">

  <div className="auth-card">

    <div className="auth-brand">Campus SMS</div>

    <div className="auth-sub">Sign in to manage students, courses and enrollments.</div>

    {error && <div className="alert alert-danger">{error}</div>}

    <form onSubmit={handleSubmit}>

      <div className="form-group">

        <label htmlFor="email">Email</label>

        <input

          id="email"

          type="email"

          className="form-control"

          value={email}

          onChange={(e) => setEmail(e.target.value)}

          required

          autoFocus

        />

      </div>

      <div className="form-group">

        <label htmlFor="password">Password</label>

        <input
```

```

      id="password"

      type="password"

      className="form-control"

      value={password}

      onChange={(e) => setPassword(e.target.value)}

      required

    />

  </div>

  <button type="submit" className="btn btn-primary btn-block" disabled={loading}>

    {loading ? 'Signing in...' : 'Sign In'}

  </button>

</form>

<div className="auth-switch">

  New here? <Link to="/register">Create an account</Link>

</div>

</div>

</div>

);

}

```

src/pages/Register.jsx

```
import { useState } from 'react';
```

```
import { Link, useNavigate } from 'react-router-dom';

import { useAuth } from '../context/AuthContext';

export default function Register() {

  const [form, setForm] = useState({ name: "", email: "", password: "", role: 'admin' });

  const [error, setError] = useState("");

  const [loading, setLoading] = useState(false);

  const { register } = useAuth();

  const navigate = useNavigate();

  const handleChange = (e) => setForm({ ...form, [e.target.name]: e.target.value });

  const handleSubmit = async (e) => {

    e.preventDefault();

    setError("");

    setLoading(true);

    try {

      await register(form);

      navigate('/');

    } catch (err) {

      setError(err.response?.data?.message || 'Registration failed.');
```

```
    } finally {

      setLoading(false);

    }

  };
}
```

```
return (  
  
  <div className="auth-wrapper">  
  
    <div className="auth-card">  
  
      <div className="auth-brand">Campus SMS</div>  
  
      <div className="auth-sub">Create a staff account to get started.</div>  
  
      {error && <div className="alert alert-danger">{error}</div>}  
  
      <form onSubmit={handleSubmit}>  
  
        <div className="form-group">  
  
          <label htmlFor="name">Full Name</label>  
  
          <input id="name" name="name" className="form-control" value={form.name}  
onChange={handleChange} required autoFocus />  
  
        </div>  
  
        <div className="form-group">  
  
          <label htmlFor="email">Email</label>  
  
          <input id="email" name="email" type="email" className="form-control"  
value={form.email} onChange={handleChange} required />  
  
        </div>  
  
        <div className="form-group">  
  
          <label htmlFor="password">Password</label>  
  
          <input  
  
            id="password"  
  
            name="password"
```

```
        type="password"

        className="form-control"

        value={form.password}

        onChange={handleChange}

        required

        minLength={6}

    />

</div>

<div className="form-group">

    <label htmlFor="role">Role</label>

    <select id="role" name="role" className="form-control" value={form.role}
onChange={handleChange}>

        <option value="admin">Admin</option>

        <option value="teacher">Teacher</option>

        <option value="student">Student</option>

    </select>

</div>

<button type="submit" className="btn btn-primary btn-block" disabled={loading}>

    {loading ? 'Creating account...' : 'Create Account'}

</button>

</form>

<div className="auth-switch">
```


Already have an account? <Link to="/login">Sign in</Link>

</div>

</div>

</div>

);

}

src/pages/Dashboard.jsx

```
import { useEffect, useState } from 'react';
```

```
import { studentService } from '../services/studentService';
```

```
import { courseService } from '../services/courseService';
```

```
import { enrollmentService } from '../services/enrollmentService';
```

```
export default function Dashboard() {
```

```
  const [stats, setStats] = useState({ students: 0, courses: 0, enrollments: 0, active: 0 });
```

```
  const [loading, setLoading] = useState(true);
```

```
  useEffect(() => {
```

```
    async function loadStats() {
```

```
      try {
```

```
        const [studentsRes, coursesRes, enrollmentsRes] = await Promise.all([
```

```
          studentService.list({ limit: 100 }),
```

```
          courseService.list(),
```

```
          enrollmentService.list()
```

```

]);

const students = studentsRes.data.students || [];

setStats({

  students: studentsRes.data.total ?? students.length,

  courses: coursesRes.data.length,

  enrollments: enrollmentsRes.data.length,

  active: students.filter((s) => s.status === 'active').length

});

} finally {

  setLoading(false);

}

}

loadStats();

}, []);

const cards = [

  { label: 'Total Students', value: stats.students },

  { label: 'Active Students', value: stats.active },

  { label: 'Courses Offered', value: stats.courses },

  { label: 'Total Enrollments', value: stats.enrollments }

];

return (

  <div>

```

```

<div className="stat-grid">

  {cards.map((c) => (

    <div className="stat-card" key={c.label}>

      <div className="stat-label">{c.label}</div>

      <div className="stat-value">{loading ? '—' : c.value}</div>

    </div>

  ))}

</div>

<div className="card">

  <div className="card-header">

    <h3>Welcome back</h3>

  </div>

  <div style={{ padding: 20, color: 'var(--color-text-muted)', fontSize: '0.9rem', lineHeight: 1.6
}}>

    Use the sidebar to manage student records, course offerings, and enrollments. This

    dashboard summarizes the current state of the system, pulled live from the API.

  </div>

</div>

</div>

);

}

```

src/pages/Students.jsx

```
import { useEffect, useState, useCallback } from 'react';

import { studentService } from '../services/studentService';

import { courseService } from '../services/courseService';

import StudentFormModal from '../components/StudentFormModal';

import ConfirmDialog from '../components/ConfirmDialog';

export default function Students() {

  const [students, setStudents] = useState([]);

  const [courses, setCourses] = useState([]);

  const [search, setSearch] = useState("");

  const [loading, setLoading] = useState(true);

  const [error, setError] = useState("");

  const [editing, setEditing] = useState(null); // student object or null

  const [showForm, setShowForm] = useState(false);

  const [toDelete, setToDelete] = useState(null);

  const loadStudents = useCallback(async (searchTerm = "") => {

    setLoading(true);

    setError("");

    try {

      const { data } = await studentService.list({ search: searchTerm, limit: 50 });

      setStudents(data.students);

    } catch {

    }

  }, [searchTerm]);

  useEffect(() => {

    loadStudents();

  }, [loadStudents]);

}
```

```
    } catch (err) {

        setError(err.response?.data?.message || 'Failed to load students');

    } finally {

        setLoading(false);

    }

}, []);

useEffect(() => {

    loadStudents();

    courseService.list().then((res) => setCourses(res.data)).catch(() => {});

}, [loadStudents]);

const handleSearch = (e) => {

    e.preventDefault();

    loadStudents(search);

};

const openCreate = () => {

    setEditing(null);

    setShowForm(true);

};

const openEdit = (student) => {

    setEditing(student);

    setShowForm(true);

};
```

```
const handleSave = async (payload) => {

  if (editing) {

    await studentService.update(editing.id, payload);

  } else {

    await studentService.create(payload);

  }

  setShowForm(false);

  loadStudents(search);

};

const handleDelete = async () => {

  await studentService.remove(toDelete.id);

  setToDelete(null);

  loadStudents(search);

};

return (

  <div>

    <div className="card">

      <div className="card-header">

        <h3>Students</h3>

        <div style={{ display: 'flex', gap: 10 }}>

          <form onSubmit={handleSearch} style={{ display: 'flex', gap: 8 }}>
```

```

<input

  className="search-input"

  placeholder="Search by name, roll no, or email"

  value={search}

  onChange={(e) => setSearch(e.target.value)}

/>

<button className="btn btn-ghost" type="submit">Search</button>

</form>

<button className="btn btn-accent" onClick={openCreate}>+ Add Student</button>

</div>

</div>

{error && <div className="alert alert-danger" style={{ margin: 20 }}>{error}</div>}

{loading ? (

  <div className="empty-state">Loading students...</div>

) : students.length === 0 ? (

  <div className="empty-state">No students found. Add your first student to get
started.</div>

) : (

  <table>

    <thead>

      <tr>

        <th>Roll No.</th>

```

```

    <th>Name</th>

    <th>Email</th>

    <th>Course</th>

    <th>Status</th>

    <th></th>

  </tr>

</thead>

<tbody>

  {students.map((s) => (

    <tr key={s.id}>

      <td><span className="roll-badge">{s.rollNumber}</span></td>

      <td>{s.name}</td>

      <td>{s.email}</td>

      <td>{s.course ? `${s.course.code}` : '—'}</td>

      <td><span className={`status-pill ${s.status}`}>{s.status}</span></td>

      <td>

        <div className="row-actions">

          <button className="btn btn-ghost btn-sm" onClick={() =>
openEdit(s)}>Edit</button>

          <button className="btn btn-danger-ghost btn-sm" onClick={() =>
setToDelete(s)}>Delete</button>

        </div>

      </td>

```



```

        </tr>

    )}}

</tbody>

</table>

)}

</div>

{showForm && (

    <StudentFormModal

        courses={courses}

        initialData={editing}

        onSave={handleSave}

        onClose={() => setShowForm(false)}

    />

)}

{toDelete && (

    <ConfirmDialog

        title="Delete student?"

        message={` This will permanently remove ${toDelete.name} (${toDelete.rollNumber}) and
their enrollment history.`}

        onConfirm={handleDelete}

        onCancel={() => setToDelete(null)}

    />

```

```
    })  
  </div>  
  
);  
}
```

src/pages/Courses.jsx

```
import { useEffect, useState, useCallback } from 'react';  
  
import { courseService } from '../services/courseService';  
  
import CourseFormModal from '../components/CourseFormModal';  
  
import ConfirmDialog from '../components/ConfirmDialog';  
  
export default function Courses() {  
  
  const [courses, setCourses] = useState([]);  
  
  const [loading, setLoading] = useState(true);  
  
  const [error, setError] = useState("");  
  
  const [editing, setEditing] = useState(null);  
  
  const [showForm, setShowForm] = useState(false);  
  
  const [toDelete, setToDelete] = useState(null);  
  
  const loadCourses = useCallback(async () => {  
  
    setLoading(true);  
  
    setError("");  
  
    try {  
  
      const { data } = await courseService.list();
```

```
    setCourses(data);

  } catch (err) {

    setError(err.response?.data?.message || 'Failed to load courses');

  } finally {

    setLoading(false);

  }

}, []);

useEffect(() => { loadCourses(); }, [loadCourses]);

const handleSave = async (payload) => {

  if (editing) {

    await courseService.update(editing.id, payload);

  } else {

    await courseService.create(payload);

  }

  setShowForm(false);

  loadCourses();

};

const handleDelete = async () => {

  await courseService.remove(toDelete.id);

  setToDelete(null);

  loadCourses();
```

```
};

return (

  <div>

    <div className="card">

      <div className="card-header">

        <h3>Courses</h3>

        <button className="btn btn-accent" onClick={() => { setEditing(null); setShowForm(true);
}}>+ Add Course</button>

      </div>

      {error && <div className="alert alert-danger" style={{ margin: 20 }}>{error}</div>}

      {loading ? (

        <div className="empty-state">Loading courses...</div>

      ) : courses.length === 0 ? (

        <div className="empty-state">No courses yet. Add the first course offering.</div>

      ) : (

        <table>

          <thead>

            <tr>

              <th>Code</th>

              <th>Name</th>

              <th>Duration</th>

              <th>Description</th>
```

```

        <th></th>

    </tr>

</thead>

<tbody>

    {courses.map((c) => (

        <tr key={c.id}>

            <td><span className="roll-badge">{c.code}</span></td>

            <td>{c.name}</td>

            <td>{c.durationMonths} months</td>

            <td style={{ color: 'var(--color-text-muted)' }}>{c.description || '—'}</td>

            <td>

                <div className="row-actions">

                    <button className="btn btn-ghost btn-sm" onClick={() => { setEditing(c);
setShowForm(true); }}>Edit</button>

                    <button className="btn btn-danger-ghost btn-sm" onClick={() =>
setToDelete(c)}>Delete</button>

                </div>

            </td>

        </tr>

    )})

</tbody>

</table>

    )}

```

```

    </div>

    {showForm && (

      <CourseFormModal initialData={editing} onSave={handleSave} onClose={() =>
setShowForm(false)} />

    )}

    {toDelete && (

      <ConfirmDialog

        title="Delete course?"

        message={`This will permanently remove ${toDelete.name} (${toDelete.code}).`}

        onConfirm={handleDelete}

        onCancel={() => setToDelete(null)}

      />

    )}

  </div>

);

}

```

src/pages/Enrollments.jsx

```

import { useEffect, useState, useCallback } from 'react';

import { enrollmentService } from '../services/enrollmentService';

import { studentService } from '../services/studentService';

import { courseService } from '../services/courseService';

```

```
import EnrollmentFormModal from '../components/EnrollmentFormModal';

import ConfirmDialog from '../components/ConfirmDialog';

export default function Enrollments() {

  const [enrollments, setEnrollments] = useState([]);

  const [students, setStudents] = useState([]);

  const [courses, setCourses] = useState([]);

  const [loading, setLoading] = useState(true);

  const [error, setError] = useState("");

  const [showForm, setShowForm] = useState(false);

  const [toDelete, setToDelete] = useState(null);

  const loadEnrollments = useCallback(async () => {

    setLoading(true);

    setError("");

    try {

      const { data } = await enrollmentService.list();

      setEnrollments(data);

    } catch (err) {

      setError(err.response?.data?.message || 'Failed to load enrollments');

    } finally {

      setLoading(false);

    }

  }, []);
```

```

useEffect(() => {

  loadEnrollments();

  studentService.list({ limit: 200 }).then((res) => setStudents(res.data.students)).catch(() => {});

  courseService.list().then((res) => setCourses(res.data)).catch(() => {});

}, [loadEnrollments]);

const handleSave = async (payload) => {

  await enrollmentService.create(payload);

  setShowForm(false);

  loadEnrollments();

};

const handleDelete = async () => {

  await enrollmentService.remove(toDelete.id);

  setToDelete(null);

  loadEnrollments();

};

return (

  <div>

    <div className="card">

      <div className="card-header">

        <h3>Enrollments</h3>

        <button className="btn btn-accent" onClick={() => setShowForm(true)}>+ New Enrollment</button>

```



```
</div>
```

```
{error && <div className="alert alert-danger" style={{ margin: 20 }}>{error}</div>}
```

```
{loading ? (
```

```
  <div className="empty-state">Loading enrollments...</div>
```

```
) : enrollments.length === 0 ? (
```

```
  <div className="empty-state">No enrollments yet. Enroll a student into a course.</div>
```

```
) : (
```

```
  <table>
```

```
    <thead>
```

```
      <tr>
```

```
        <th>Student</th>
```

```
        <th>Course</th>
```

```
        <th>Enrollment Date</th>
```

```
        <th>Grade</th>
```

```
        <th>Status</th>
```

```
        <th></th>
```

```
      </tr>
```

```
    </thead>
```

```
    <tbody>
```

```
      {enrollments.map((e) => (
```

```
        <tr key={e.id}>
```

```

        <td>{e.student?.name} <span
className="roll-badge">{e.student?.rollNumber}</span></td>

        <td>{e.course?.code} — {e.course?.name}</td>

        <td>{e.enrollmentDate}</td>

        <td>{e.grade || '—'}</td>

        <td><span className={`status-pill ${e.status === 'enrolled' ? 'active' : e.status ===
'completed' ? 'graduated' : 'inactive'}`}>{e.status}</span></td>

        <td>

        <div className="row-actions">

            <button className="btn btn-danger-ghost btn-sm" onClick={() =>
setToDelete(e)}>Remove</button>

        </div>

        </td>

    </tr>

    )}}

</tbody>

</table>

)}

</div>

{showForm && (

    <EnrollmentFormModal

        students={students}

        courses={courses}

```

```
      onSave={handleSave}

      onClose={() => setShowForm(false)}

    />

  )}

  {toDelete && (

    <ConfirmDialog

      title="Remove enrollment?"

      message={`This will unenroll ${toDelete.student?.name} from
${toDelete.course?.name}.`}

      onConfirm={handleDelete}

      onCancel={() => setToDelete(null)}

    />

  )}

</div>

);

}
```